

Impact of Ai on Students

About the Dataset: As Generative AI tools like ChatGPT, Copilot, and Gemini become deeply embedded in academic life, understanding their **true impact on students** has never been more critical. This dataset provides a rich, multi-dimensional snapshot of student interactions with AI — spanning academic outcomes, behavioral patterns, institutional context, and psychological well-being.

This dataset captures the impact of Generative AI tools on students across various academic disciplines. It tracks how students use AI (like ChatGPT, Copilot, etc.), their study habits, academic performance before and after a semester, and their mental well-being — giving a 360° view of AI's role in modern education.

Data Processing Steps:

1. Imported the required Python libraries (Pandas and NumPy).
2. Loaded the dataset into a Pandas Data Frame.
3. Displayed the first and last few records to inspect the data.
4. Checked the dataset shape, column names, and data types.
5. Identified missing values using `is null (). sum ()`.
6. Checked for duplicate records using `duplicated (). sum ()`.
7. Removed duplicate rows (if any).
8. Generated descriptive statistics using `describe ()`.
9. Verified data consistency and prepared the dataset for analysis.
10. Saved the cleaned dataset for further processing and model training.

Statistical Summary: Generate summary statistics for numerical features.

Data Validation: Verify numerical columns.

- Verify categorical columns. - Ensure data consistency.

Save Processed Dataset: Store the cleaned dataset for further analysis.

Dataset Overview:

Property	Details
Rows	50,000 students
Columns	16 features
Missing Values	None (fully complete dataset)
File Format	CSV
Target Variables	Post_Semester_GPA, Skill_Retention_Score, Burnout_Risk_Level

Potential Use Cases

This dataset supports a wide range of analysis and modelling tasks:

-  **GPA Change Analysis** – How does AI usage correlate with academic improvement or decline?
-  **Burnout Prediction** – Build a classifier to predict Burnout_Risk_Level from study and AI habits.
-  **Skill Retention Modelling** – Does heavy AI reliance hurt long-term knowledge retention?
-  **Policy Impact Study** – Compare student outcomes under different institutional AI policies.
-  **Mental Health Insights** – Explore links between AI dependency, anxiety, and burnout.
-  **Major & Year-wise Trends** – Which disciplines benefit most (or suffer most) from AI adoption?

Feature Categories briefly:

 AI_Student_Impact_Dataset

└─  Identifier → Student_ID

└─  Academic Profile → Major Category, Year_of_Study, Pre/Post GPA

└─ 🤖 AI Behavior → Weekly_GenAI_Hours, Primary_Use_Case,
| Prompt_Engineering_Skill, Tool Diversity,
| Paid Subscription
└─ 📖 Study Habits → Traditional_Study_Hours, Perceived_AI_Dependency
└─ 🏛️ Institutional → Institutional Policy
└─ 🧠 Well-being → Anxiety_Level, Skill_Retention_Score,
Burnout_Risk_Level

Data Quality:

- ☒ **No missing values** across all 50,000 rows and 16 columns
- ☒ **Consistent encoding** for all categorical variables
- ☒ **Balanced representation** across majors, year levels, and policies

License:

Please check the dataset license before using it. Ensure appropriate citation if used in academic or commercial projects.

Acknowledgements:

If you use this dataset in your work, please consider upvoting and leaving a comment describing your analysis. Community feedback helps improve future versions of this dataset.